MANAS GANESH DASARI

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Portfolio | GitHub | LinkedIn

PROFILE

Computer Science undergraduate (2028) interested in neural architectures and model generalization. Builds reproducible NLP and EEG research pipelines, evaluates models against meaningful baselines, and develops practical engineering tools. Seeking AI/ML engineering and research internships.

EDUCATION

Amrita School of Engineering, Bengaluru | B.Tech in Computer Science | Expected 2028
Relevant coursework: Machine Learning, Neural Networks & Deep Learning, Probability & Statistics, Data Structures & Algorithms

TECHNICAL SKILLS

Programming: Python, Go, Java, JavaScript/TypeScript, SQL, Bash
ML & Research: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, MNE-Python, SHAP, statsmodels
Methods & Tools: Model training, feature engineering, cross-validation, experiment design, Git, Linux, Docker, FastAPI

MACHINE LEARNING & RESEARCH

Clause Risk Review — NLP Model Evaluation | Python · PyTorch · Transformers · scikit-learn

- Compared TextCNN, BERT and Legal-BERT for six-label contract risk classification on CUAD (510 annotated contracts) using controlled model-evaluation pipelines.
- Prevented overlapping-window data leakage with contract-level splits; tuned per-label thresholds on validation data and aggregated chunks using noisy-OR pooling.
- Achieved 0.879 document-level F1 / 0.933 average precision with BERT on 76 held-out contracts; found no clear Legal-BERT F1 gain and examined class-level errors and run variance.

Med-EEG / NeuroDial — Interpretable Biosignal ML | Python · MNE · scikit-learn · SHAP

- Built a reproducible EEG pipeline across two OpenNeuro datasets, using MNE preprocessing, ICA artifact control, spectral features and 64-channel covariance representations.
- Decoded meditation depth from 1,707 reports by 39 participants with retrospective subject-wise evaluation: Riemannian model AUC 0.609 (Holm p = 0.035), +0.091 over the strongest non-EEG baseline.
- Pre-registered analysis detected depth-related EEG signal 60–120 s before reports (AUC 0.584; Holm p = 0.020); documented failed replication of the earlier gamma/theta hypothesis.

ENGINEERING & OPTIMIZATION

Sprout — AI-Oriented Codebase Intelligence CLI | Go · Git · dependency graphs · MCP

- Engineered a standard-library Go CLI that maps multi-language repositories, identifies file dependencies and change impact, and recommends relevant tests.
- Exposed repository maps and impact analysis as MCP tools for coding agents; shipped installable releases through Homebrew and Linux package formats.

Atmos — Urban Sensor Placement Optimization | Python · NumPy · FastAPI

- Modelled Bengaluru sensor placement using live WAQI inputs, traffic-weighted zones, bounding-phase search and Golden Ratio Search to explore coverage–cost trade-offs.

Flow — Real-Time Collaborative Study Platform | TypeScript · Astro · Node.js · WebRTC · PostgreSQL

- Built collaborative study rooms with WebRTC calls, shared focus timers, room tasks and persistent sessions; integrated Supabase authentication and Realtime.
- Refactored the interface into Astro and strict TypeScript; added server, SQL and browser checks in CI.

OPEN SOURCE

Exercism — VB.NET Track Contributor | **12 merged pull requests**

- Updated exercise test suites and synchronized problem specifications across linked-list, tree-building, banking and other programming exercises.

Gleam — Contributor | **merged pull request**

- Improved dependency-update navigation by linking changed packages to Hex diffs.

TECHNICAL WRITING

15-chapter ML experiment handbook | **Building Sprout: engineering case study**